

Enhanced electrical properties of NaBi modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ -based Aurivillius piezoceramics via structural distortion

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ABSTRACT

Two-layer structured ferroelectric ceramics, $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ($x = 0.0\text{--}0.5$; CNBN- x), were prepared by solid-state sintering. An artificial pseudo-morphotropic phase boundary was formed by the substitution of Ca with NaBi, and the crystal structure tended to crystallise in the pseudo-tetragonal phase as the increase of NaBi content. Curie temperature (T_c) slightly reduce and remain high value with the increase of NaBi content, which is in the range of 880–935 °C. Moreover, enhanced piezoactivity (d_{33}) and ferroelectricity (P_r) accompanying with decreased dielectric loss can be acquired in NaBi-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$. The CNBN-0.1 had the best performance, and its T_c , d_{33} and P_r was 925 °C, 14.4 pC/N and 6.0 $\mu\text{C}/\text{cm}^2$, respectively. Thermal annealing measurement exhibited that the piezoelectric properties of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ had good temperature stability up to 800 °C. Therefore, the NaBi-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ ceramics are perceived to be of promising candidates for high-temperature applications.

1. Introduction

Piezoelectric sensors at high temperature such as accelerometers, high temperature SAW (Surface Acoustic Wave) sensors, acoustic emission sensors and gas/pressure sensors, etc. are widely used to detect and record dynamic mechanical parameters at harsh conditions for structural control and health monitoring system in the fields of aircraft, aerospace and nuclear power industries [1–4]. Take the aircraft and aerospace industry as an example, the piezoelectric accelerometers are usually required to be the closest approach to the engine components for sufficient detection sensitivity to effectively monitor the working situation of the engine, which demands devices can endure high temperatures more than 500 °C [4]. This indicates that the piezoelectric materials must be able to function well and have very good thermal stability in a wide temperature range since they are the key component of high-temperature piezoelectric sensors.

From the viewpoint of piezoelectric sensors, the maximum operation temperature is restricted by both Curie temperature (T_c) and sensitivity level, which is settled by the piezoelectric coefficients, the electrical resistivity and the dielectric losses of piezoelectric materials [5]. Aurivillius-type piezoceramics, with the general chemical formula of $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$, are excellent candidates for high

temperature sensor applications due to their high T_c . Since the layers of $m(\text{ABO}_3)^{2-}$ alternate with $(\text{Bi}_2\text{O}_2)^{2+}$ layers, however, the rotation of their spontaneous polarisation (P_s) is limited in two-dimensional orientation, leading to relatively low piezoactivity [6]. Another concern for is their higher conductivity at elevated temperature [7,8], which conduces to charge drift disturbing the formation of piezoelectrically induced charges, thereby the sensitivity level and dependability of piezoelectric sensors was reduced remarkably [4].

Two-layer Aurivillius phase piezoelectric ceramics, $\text{MBi}_2\text{Nb}_2\text{O}_9$ (M represents the A-site cation), have attracted significant research interests from the technological and scientific points of view due to their unique overall performance. Among them, $\text{CaBi}_2\text{Nb}_2\text{O}_9$ (CBN, $m = 2$) is prospective for high-temperature piezoelectric sensor applications because that it has the highest-known T_c of 943 °C [9]. However, the piezoelectric coefficient (d_{33}) of pure CBN ceramic is relatively low (~ 6 pC/N) [10]. Meanwhile, $\text{Na}_{0.5}\text{Bi}_{2.5}\text{Nb}_2\text{O}_9$ (NBN, $m = 2$) is also a typical two-layered Aurivillius phase ferroelectric ceramics with a high T_c (790 °C), but relatively low piezoelectric coefficient (~ 6 pC/N) [11]. Many approaches have been carried out to enhance the piezoelectric properties of CBN and NBN ceramics via texture techniques (such as spark plasma sintering [9], templated grain growth [12], hot pressing [13] etc.) and/or cation doping modifications [14,15]. It was found that

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the d_{33} of CBN and NBN ceramics can be clearly improved by chemical modification (multi-rare-earth dopants) in A-site cations. Nevertheless, there is little attention has been paid to enhance the electric properties of BLSFs in the viewpoint of the construction of an artificial morphotropic phase boundary. The construction of phase boundaries is an effective method of the enhancement of the piezoelectrical performance in many materials [16,17]. It is reported that the piezoelectric performance in (K,Na)NbO₃-based ceramics can be improved through the method of manipulating the rhombohedral-tetragonal (R-T) phase boundary [18,19].

Therefore, in the contribution, we designed a new two-layered Aurivillius-type compound Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ by the introduction of A site cations of the typical two-layer Aurivillius-type compound NBN. The detailed crystalline structure variation of NaBi-modified CBN ceramics was analysed. The relationship between the structure variation and the electrical properties (dielectric property, ferroelectricity and piezoelectrical property) was investigated systematically.

2. Experimental procedure

Ceramic samples with compositions of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ($x = 0.0, 0.1, 0.3$, and 0.5 , abbreviated as CBN, CNBN-0.1, CNBN-0.3 and CNBN-0.5, respectively) were prepared using the solid-state sintering. The starting raw materials are CaCO₃ (99.00%), Na₂CO₃ (98.92%), Bi₂O₃ (99.78%) and Nb₂O₅ (99.95%). All raw materials were weighed in the stoichiometric ratio and planetary milling for 4 h. Calcination of dried powders was performed at 850 °C for 2 h after which ball-milling again for 4 h under the same conditions was carried out. The dried powders with 7 wt% binder (polyvinyl alcohol) were pressed into pellets. The pellets were maintained at 750 °C for 2 h to remove the binder. All samples were sintered at 1100 °C for 2 h and then processed into the same size (12 mm in diameter, 0.5 mm in thickness) for performance measurements.

X-ray diffraction (XRD) patterns of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics were obtained by X-ray diffractometer (D/max2550V, Rigaku, Co., Tokyo, Japan) with Cu K α radiation. Natural microstructures were examined by using scanning electron microscopy (SEM; Hitachi, Tokyo, Japan). Raman spectra were measured by using micro-Raman spectrometer (Jobin-Yvon Lab RAW HR 800 UV) equipped with the exciting source with the 632.8 nm line from He-Ne laser. In order to measure electrical properties, the platinum paste was coated on the two surfaces of the samples. Dielectric properties were characterised by E4980A LCR meter. Ferroelectric properties measurements were performed by using TF-2000 ferroelectric analyser (aixACCT, Aachen, Germany). DC electrical resistivity measurement was carried out by using the resistivity test system HRMS-900 (Partulab, Wuhan, China) in the range of 200–700 °C. All ceramics samples were polarised by applying different dc electric fields for 20 min in silicone oil bath of 200 °C. The d_{33} were measured by a quasi-static ZJ-6 d_{33} meter (Institute of Acoustics, Chinese Academy of Sciences, China). Temperature stability of piezoelectric properties of all the compositions was achieved by post-annealing the polarised samples at different temperatures for 4 h, and re-measuring d_{33} when the samples were cooled to room temperature. The above procedure was repeated at various temperatures until 950 °C.

3. Results and discussion

XRD patterns of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics are shown in Fig. 1(a). All the obvious diffraction peaks match the main phase of CBN on the basis of the JCPDS 49–0608, implying that additive (NaBi) have been entered into the solid solution with the CBN as designed. All the XRD patterns of the samples present the strongest diffraction peaks in the (115) plane, which agrees with the (112 $\overline{m}$ +1) most intense reflection peak for BLSFs compounds. Moreover, the expanded XRD patterns in the range of 32.5–33.5° and 33.4–34.0° are shown in Fig. 1(b) and (c). It is interesting to note that the diffraction peaks of the

(200)/(020) and (202)/(022) planes gradually merge into one peak with increasing NaBi content respectively, manifesting that the presence of a change in the crystal structure in the vicinity of composition with $x = 0.1$. As we know, the ferroelectric phase of CBN is the orthorhombic system, while its paraelectric phase belongs to tetragonality [20]. Therefore, there may be a structure transformation between orthorhombic symmetry and pseudo-tetragonal symmetry nearby $x = 0.1$.

To further analyse the crystal structure change of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics, Rietveld refinement analysis was performed for samples with $x = 0.1$ and 0.3 using the A2₁am space group at room temperature. The Rietveld refined XRD patterns of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics are shown in Fig. 2. The refined results are in line with experimental data with reasonable reliability factors (R_{wp} , R_p and χ^2). Crystal data and structure refinement conditions for CNBN-0.1 and CNBN-0.3 are listed in the Table 1. The a/b ratio is reduced from 1.00722 to 1.00569 with the x increasing, which further demonstrates that the orthorhombicity is weakened.

Fig. 3 shows the natural microstructures of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics. It was found that each composition displays platelike grains with randomly oriented because of the higher grain growth rate in the a - b plane of pseudo-perovskites. The NaBi-modified CBN ceramics possess the typical structure of Aurivillius phase ferroelectric ceramics, which is in agreement with the main phase of CBN by the XRD analysis. In addition, samples with different x amount present well-densified natural microstructures meanwhile there are no obvious defects (pores and impurity), further verifying the fact that Na and Bi had entered into the matrix lattice well. Moreover, the relative densities of were measured to be 94.4%, 94.8%, 95.2% and 94.6% for CBN, CNBN-0.1, CNBN-0.3 and CNBN-0.5 by the Archimedes method, respectively. Therefore, the change of microstructure is limited due to the introduction of NaBi additives.

Raman spectroscopy was found to be a highly effective tool in providing information on local structure change, especially detecting the variation of local order owing to atomic displacement [21]. Fig. 4 shows Raman spectra of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics in the range of 100–1000 cm⁻¹. Raman spectra of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ display the intense phonon modes at 187, 210, 298, 327, 451, 596 and 822 cm⁻¹, which is similar to those of SrBi₂Nb₂O₉ [22]. The Raman spectra can be segregated into thirteen individual components by Lorentz fitting. The modes at 187 cm⁻¹, 210 cm⁻¹ and 234 cm⁻¹ are dominated by the stretching vibrations (A-O1) and bending vibrations (A-O1-A) [23,24]. As displayed in Fig. 4(b), the above three modes show clearly shift toward lower frequency as the increase of NaBi content. Additionally, the mode at 234 cm⁻¹ gradually becomes weaker and finally disappears in the $x = 0.5$ component. It implies that the NaBi doping occupies the A-site and induces the change of vibrational frequencies. This result is consistent with the XRD patterns analysis results. The vibrations of the NbO₆ octahedron prevailing in the higher frequencies 822 cm⁻¹ and 846 cm⁻¹ [25] exhibit a visible low-frequency shift, as shown in Fig. 4(c). The result reveals that the NaBi substitution for Ca would be effect on the higher Raman vibration modes. This may be attributed to the replacement of Ca ions by smaller radius ions NaBi ($R(\text{Ca}^{2+}) = 1.34 \text{ \AA}$, $R(\text{Na}^{+}) = 1.02 \text{ \AA}$, $R(\text{Bi}^{3+}) = 1.35 \text{ \AA}$, $R_{\text{NB}} = 0.5 * (1.02 + 1.35) = 1.185 \text{ \AA}$, R_{NB} represents the ionic radius of (NaBi)_{0.5}²⁺ [26], resulting in an enlarged rattling space of NbO₆ octahedron.

Temperature dependence of dielectric constant (ϵ) and dielectric loss ($\tan\delta$) of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics in the range of 25 – 950 °C at 1 MHz, as displayed in Fig. 5. It can be seen that all components have evident dielectric peak corresponding to Curie temperature (T_c), which indicates the phase transition from ferroelectric state to paraelectric state. The T_c of pure CBN ceramics is 935 °C, which is identical to reports from other researchers [27,28]. The T_c values of Ca_{1-x}(NaBi)_{0.5x}Bi₂Nb₂O₉ ceramics slightly decrease from 925 °C to 880 °C as the content of NaBi increase. Besides, the dielectric loss ($\tan\delta$)

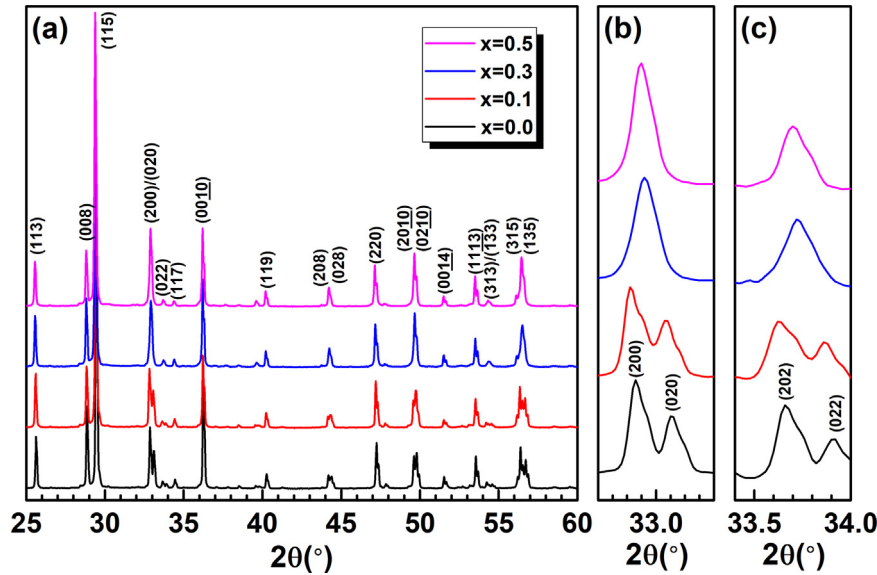


Fig. 1. (a) XRD patterns, (b) and (c) explain XRD patterns of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics.

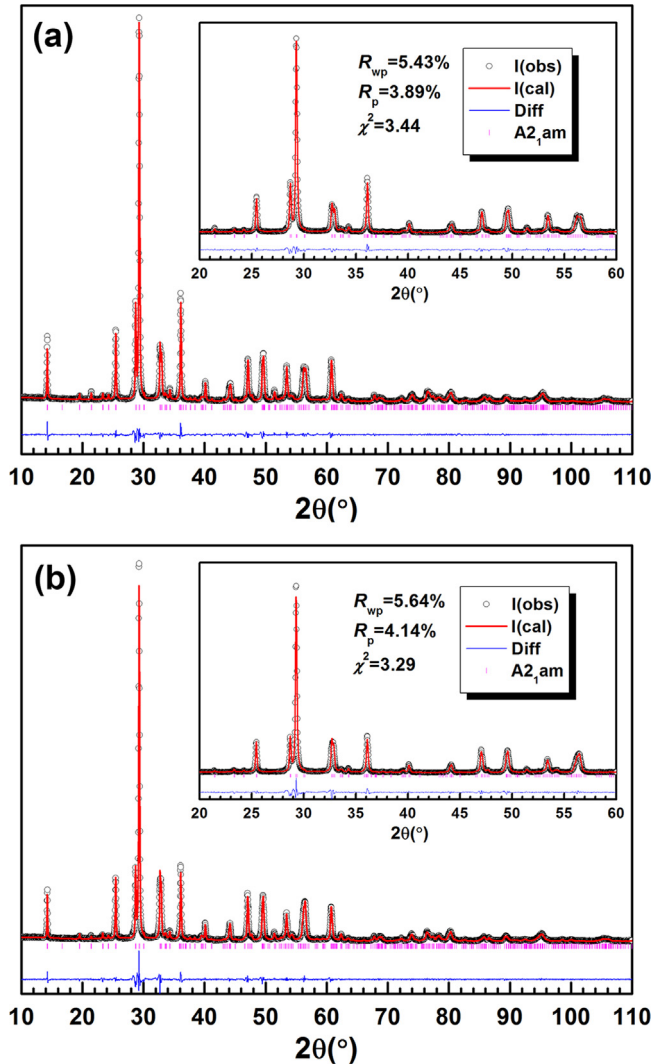


Fig. 2. Rietveld refinement for $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics: (a) CNBN-0.1 and (b) CNBN-0.3.

Table 1

Crystal data and structure refinement conditions of CNBN-0.1 and CNBN-0.3.

Chemical formula	$\text{Ca}_{0.9}(\text{Na}_{0.05}\text{Bi}_{2.05})\text{Nb}_2\text{O}_9$	$\text{Ca}_{0.7}(\text{Na}_{0.15}\text{Bi}_{2.15})\text{Nb}_2\text{O}_9$
Crystal structure	Orthorhombic	Orthorhombic
Space group	$A2_1am$	$A2_1am$
a (Å)	5.47727 (35)	5.47663(38)
b (Å)	5.43798(35)	5.44562(37)
c (Å)	24.86545(25)	24.87476(40)
V (Å ³)	740.625 (39)	741.858(40)
Crystal density (g/cm ³)	7.134	7.258
Reduced χ^2	3.44	3.29
R_{wp} (%)	5.43	5.64
R_p (%)	3.89	4.14
a/b	1.00722(51)	1.00569(45)

of pure CBN increase linearly above 300 °C, while the $\tan\delta$ of NaBi-modified CBN ceramics keeps a very smaller value with the temperature up to 500 °C. The higher T_c and lower $\tan\delta$ of NaBi-modified CBN ceramics are beneficial to the application of the high-temperature.

Fig. 6 illustrates the polarisation versus electric field (P - E) and current versus electric field (I - E) loops of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics measured at 1 Hz and 180 °C. It is observed that NaBi-modified CBN ceramics display a more typical and saturated ferroelectric P - E loops, compared with the results reported by other research groups on CBN-based ceramics [29]. The reversal of spontaneous polarisation emerges near the coercive field (E_c) and leads to a current peak (I_{max}) in the I - E loops [30]. It is generally believed that switching current peak is a more accurate means to watch the switching behaviour of polarisation, and evaluates the ferroelectric properties of materials [31]. As it can be seen, the current peaks become sharper with the increase of x content, suggesting that more ferroelectric domains switching and the enhancement of ferroelectricity and $x = 0.1$ composition possesses the good ferroelectric property with P_r of 6.0 $\mu\text{C}/\text{cm}^2$.

The temperature dependence of DC resistivity of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics is shown in Fig. 7. It was found that the introduction of NaBi has little effect on the DC resistivity of CBN ceramics. The DC resistivity of samples is greater than $1 \times 10^{12} \Omega\cdot\text{cm}$ at 200 °C and slightly increase with the increase of NaBi content. It can also be observed that the DC resistivity of all samples has a trend of consolidation at the higher temperature and this is especially true for the temperature above 500 °C. The larger resistivity is very important for high-temperature applications.

Fig. 8 depicts the thermal depoling behaviour of Ca_{1-x}

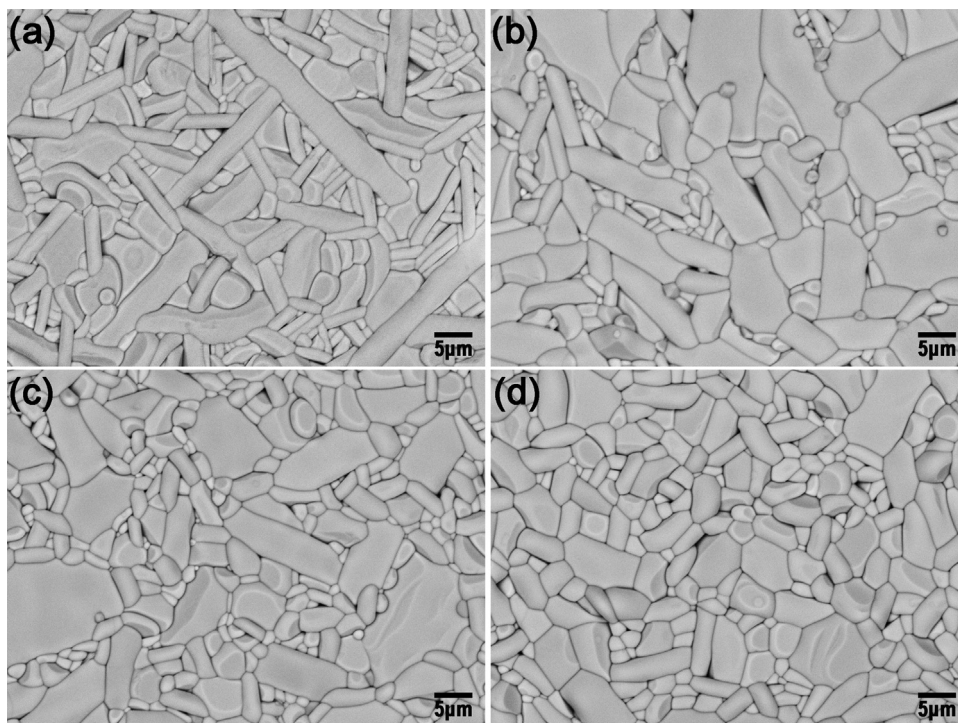


Fig. 3. Natural microstructures of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics: (a) CBN, (b) CNBN-0.1, (c) CNBN-0.3, (d) CNBN-0.5.

$\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics. The d_{33} value increased first and then decreased due to the introduction of NaBi additive at room temperature, and the maximum d_{33} was 14.4 pC/N in the composition of $x = 0.1$. Therefore, the piezoelectric properties of CBN ceramics can be promoted by the proper NaBi substitution for Ca, as compared with pure CBN ceramics ($d_{33} = 7$ pC/N). In addition, the d_{33} value remains in the order of 81% (CBN), 96% (CNBN-0.1), 95% (CNBN-0.3) and 89% (CNBN-0.5) over the temperature from room temperature to 800 °C, indicating all compositions exhibit excellent thermal stability up to higher temperature (800 °C). The result can be ascribed to the fact that

there are few non-180° domain walls for the even-layer Aurivillius compounds [32]. It is reported that non-180° domain walls are thermal instability below the T_c and result in thermal depolarisation of d_{33} [33]. The d_{33} values decrease rapidly when the annealing temperature is close to T_c , the result implies that NaBi-modified CBN ceramics possess excellent thermal stability, which is conducive to high-temperature applications.

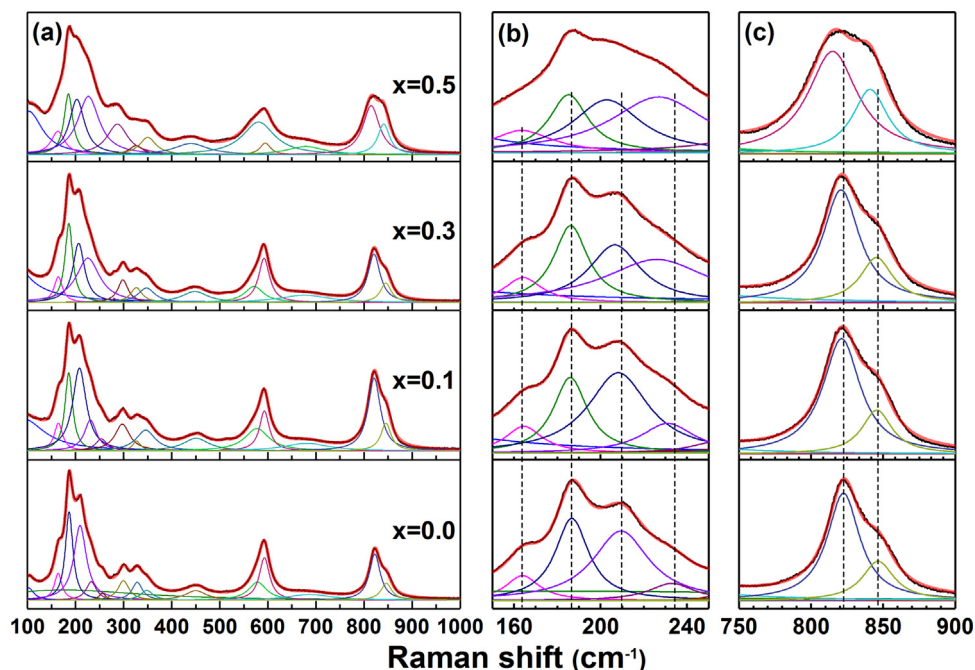


Fig. 4. Raman spectra of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ samples.

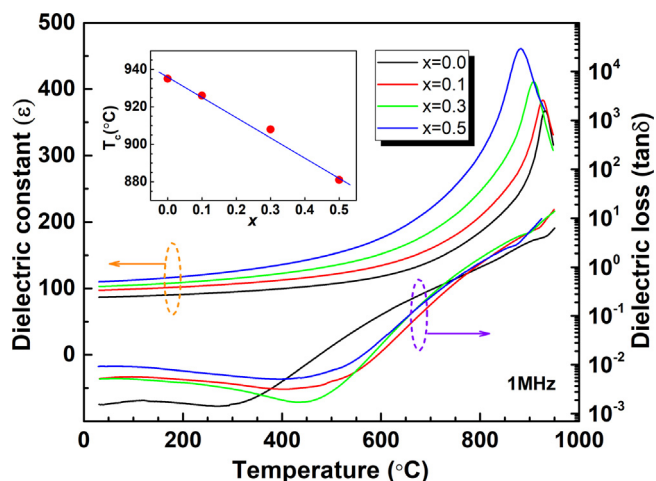


Fig. 5. The variation of dielectric constant and loss of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics as a function of temperature.

4. Conclusions

NaBi-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ Aurivillius ceramics were prepared using the conventional solid-state sintering method. The structures and microstructures of the $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ were analysed by XRD, SEM and Raman spectra. NaBi modification induced a pseudo-morphotropic phase transition from orthorhombic state to pseudo-tetragonal state in CBN ceramics. This pseudo-morphotropic phase

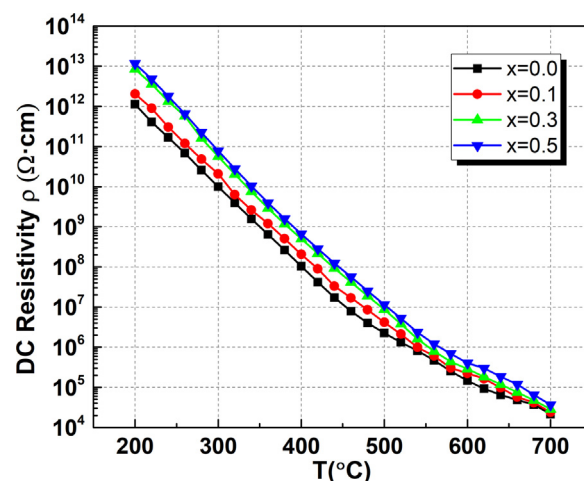


Fig. 7. DC resistivity versus temperature in $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics.

boundary can significantly improve electrical properties. NaBi-modified $\text{CaBi}_2\text{Nb}_2\text{O}_9$ exhibited superior piezoelectric properties compared with pure CBN. The optimum d_{33} of 14.4 pC/N can be obtained in $\text{Ca}_{0.9}(\text{NaBi})_{0.05}\text{Bi}_2\text{Nb}_2\text{O}_9$, which is more than double that of pure CBN composition. Therefore, $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics is a good lead-free piezoelectrical material with high T_c , stable d_{33} and high thermal-depoling temperature, which make it become a promising candidate in the field of high-temperature (over 500 °C) piezoelectric devices.

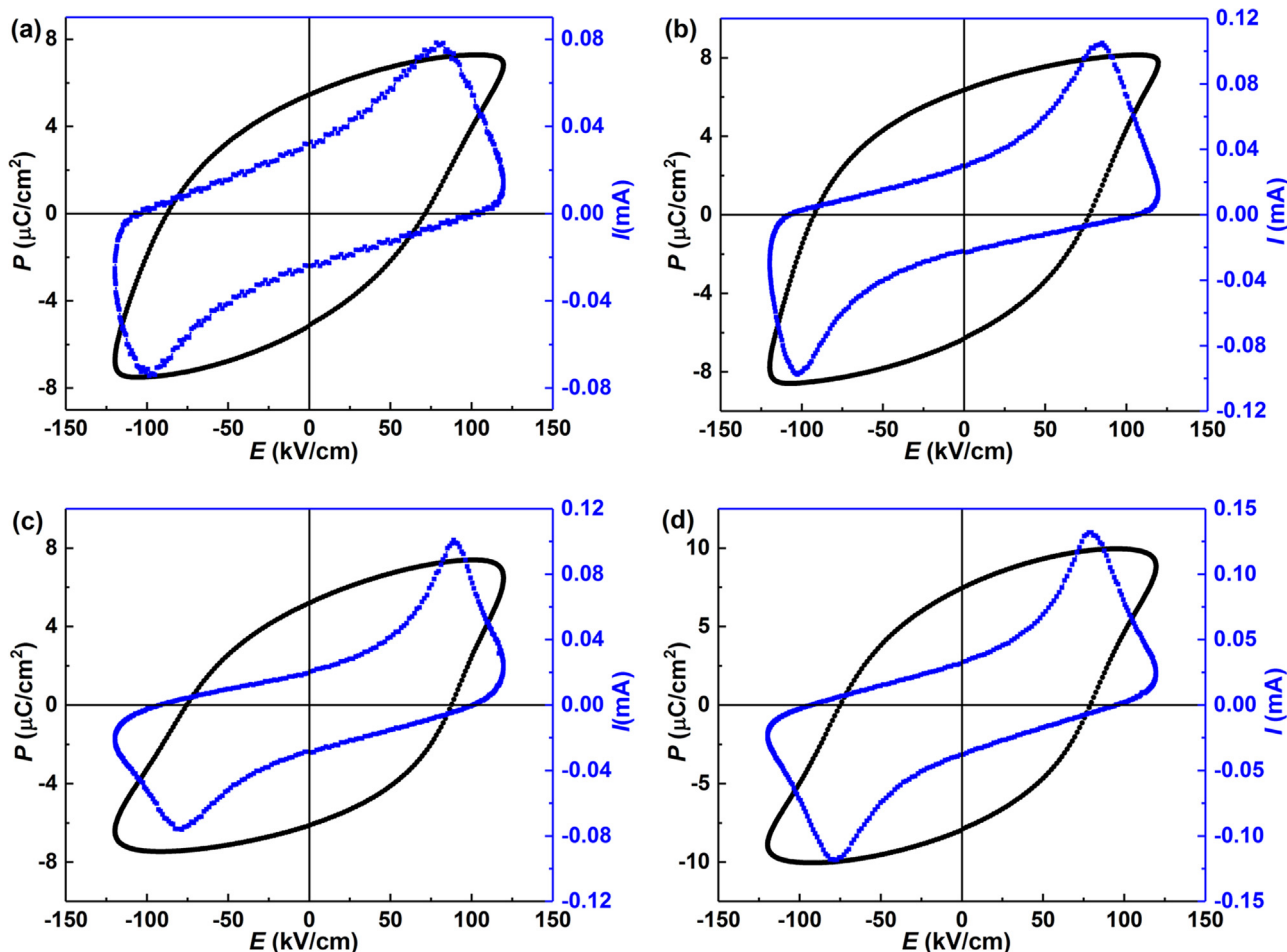


Fig. 6. Polarisation versus electric field (P - E) and current versus electric field (I - E) loops for $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics.

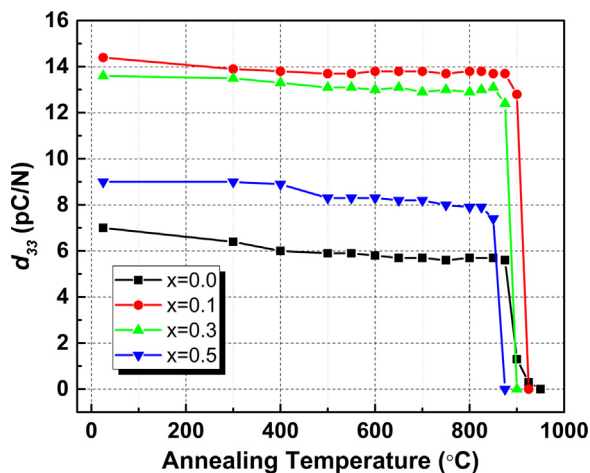


Fig. 8. The thermal depoling behaviour of $\text{Ca}_{1-x}(\text{NaBi})_{0.5x}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics.

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